What My Liked Songs Say About My Music Taste

This report analyzes my full Spotify Liked Songs library (1,814 tracks), which includes each song's audio features (energy, valence, danceability, acousticness), Spotify Popularity score, genre tags, release date, and the date I liked it. Beyond describing what I listen to, the goal is to test a diagnostic question: does the sound of a song or how old it was when I liked it predict how mainstream (popular) it is? In other words, do I gravitate toward songs because they sound like hits, or is my taste driven by something audio features can't capture?

Descriptive Overview

My liking activity is highly bursty rather than steady (Figure 1): nearly a quarter of all 1,814 songs were added in just four months (Nov 2024 – Feb 2025), suggesting periods of active music discovery followed by long stretches of passive listening. Genre-wise (Figure 2), my library is anchored in *dream pop* and *rap/cloud rap/rage rap* variants, with reggaetón and retro soul rounding out an eclectic top 10. On average, my liked tracks score 0.60 on energy, 0.58 on danceability, and 0.42 on valence (i.e., moderately upbeat sounding but not distinctly "happy" on Spotify's scale).

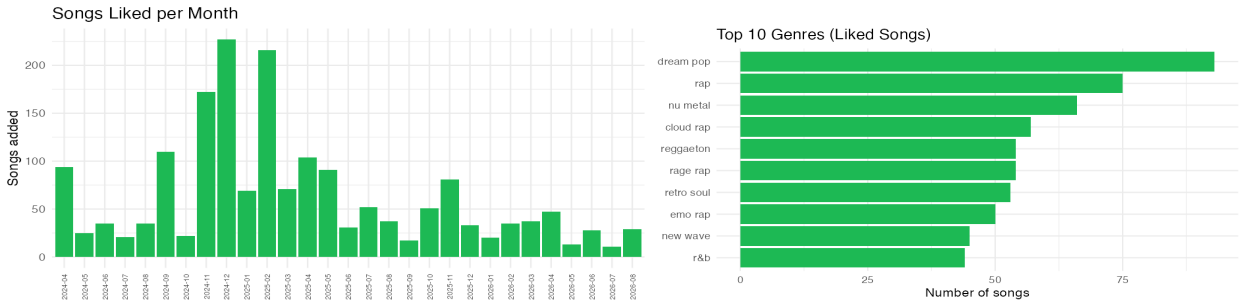


Figure 1 (left): songs liked per month April 2024 – August 2026. Figure 2 (right): top 10 primary genres by track count.

Note: 505 songs (27.8%) had no genre tag in the export and were excluded from the genre breakdown in Figure 2; this reflects gaps in Spotify's metadata rather than missing data collection on my part.

Diagnostic Analysis: What Predicts a Liked Song's Popularity?

I regressed each song's Spotify popularity score (0-100) on its audio profile and its age at the time I liked it (release date subtracted from the date added):

Predictor	Coefficient	Std. Error	P - Value
Intercept	54.72	3.17	< 0.001
Song Age When liked (yrs)	0.135	0.035	< 0.001
Energy	2.03	3.28	0.537
Danceability	4.68	3.22	0.146
Valence	-3.89	2.36	0.100

Acousticness	-1.50	2.23	0.500

N = 1,814. R^2 = 0.009; F(5,1808) = 3.31. p = 0.006

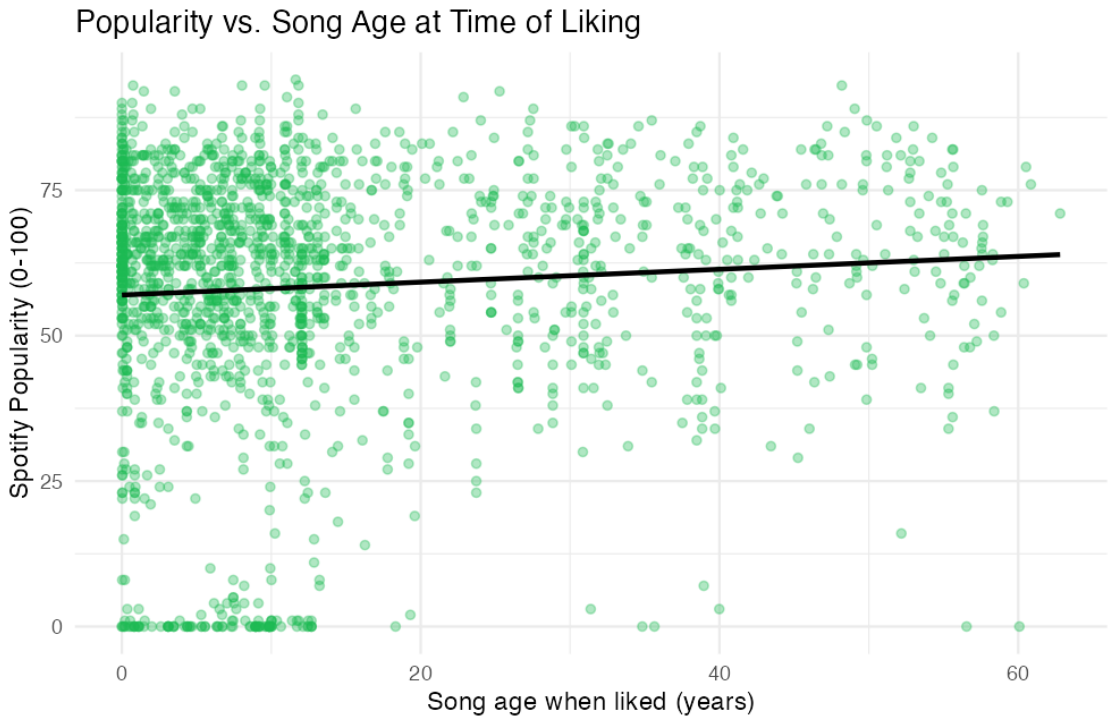


Figure 3: Popularity vs. song age at time of liking, with OLS fit line

The only statistically significant predictor is song age at the time I liked it, where each additional year of "lag" between a song's release and my liking it is associated with a 0.135-point higher Popularity score ($p < 0.001$). This is counterintuitive at first, as older finds turn out to be more mainstream, not more obscure. The likely explanation is that I tend to revisit canonized classics that have stayed popular over time, while my newest adds skew toward songs I'm discovering closer to release. Energy, danceability, valence, and acousticness are not significant predictors on their own. Most tellingly, the model's R^2 is just 0.009 even with a statistically significant F-test; these five variables explain under 1% of the variation in popularity. The diagnostic takeaway is that my music taste isn't well explained by surface-level audio traits or by chasing mainstream sound. Artist loyalty, genre scenes, and my mood during those times when I browse for music are doing the actual work and would be a natural next variable to bring in.